MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
MG5SECCHE300 - Analytical Techniques and Chemistry of Cosmetics
MODEL QUESTION PAPER
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part – A

Short answer type question
Answer any 10 questions. Each carry 2 marks

1. What are chelating agents? Give two examples. (K, CO1)
2. What are the major sources of water contamination? (K, CO1)
3. Define sampling. Explain its importance in analytical chemistry. (U, CO1)
4. How can a chemist test if a soil sample is acidic? Name one method and what result shows acidity. (A, CO1)
5. State the principle of adsorption chromatography. (K, CO2)
6. Why is an inert gas used as the mobile phase in Gas chromatography? (U, CO2)
7. Define reverse phase HPLC. (K, CO2)
8. A student removes a TLC plate from the chamber. What should the student mark immediately to determine the Rf value? (A, CO2)
9. List any four examples of cosmetics. (K, CO3)
10. Give the classification of cosmetics based on their use. (K, CO3)
11. Mention any four categories of cosmetic products. (K, CO3)
12. What is the significance of cosmetics? (U, CO3)

(10 x 2 = 20)

Essay type question
Answer any 6 questions. Each carry 5 marks

13. A student gets analytical results that are close to one another but far from the true value. Identify whether the results show good accuracy or precision and justify your answer. (A, CO1)

14. Explain any two methods used for the purification of water. (U, CO1)
15. Explain the method used for the estimation of calcium and magnesium ions in soil. (U, CO1)
16. A laboratory needs to separate volatile compounds using gas chromatography. Based on the nature of the stationary phase, identify whether gas-liquid chromatography or gas-solid chromatography would be appropriate and differentiate the two techniques. (A, CO2)
17. Explain the step-by-step procedure of TLC with the help of a labelled diagram. (U, CO2)
18. Describe the various quality-control tests carried out on cosmetics. (U, CO3)
19. Explain the physical, chemical and microbiological quality-control tests of cosmetics. (U, CO3)
20. Explain the classification of cosmetics based on their area of application, with examples. (U, CO3)

(6 x 5 = 30)
